

Arghya Datta, PhD

Applied Scientist
Worldwide Amazon Stores (WWAS)
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Educational Qualification

- August 2021 PhD in Computer Science, [Washington University in St. Louis](#), Department of Computer Science & Engineering, St. Louis, USA
- August 2021 MS in Computer Science, [Washington University in St. Louis](#), Department of Computer Science & Engineering, St. Louis, USA
- May 2015 Bachelor of Technology, [West Bengal University of Technology](#), Department of Electronics & Communication Engineering, Kolkata, India

Work Experience

- Company [Amazon.com Services LLC](#)
- Tenure Sept 2021- Present
- Role **Applied Scientist, Applied AI (e-Commerce Foundation), WorldWide Retail Stores**
- Work Summary Lead scientist for building **AgentZ**, an agentic AI framework for automating knowledge operations work across worldwide retail stores with an **annualized \$200MM in savings**
- Led the conceptualization and development of debugging agents and an systematic error taxonomy framework to analyze, classify, debug, and resolve failures in production Agentic AI systems, enabling structured diagnosis of failure modes, targeted model improvements, and measurable gains in system robustness, reliability, and interpretability. **Reduced customer tickets by 50% with automated fix deployments in Agent workflows**
 - Led R&D of **AgentZ's AI tools** such as browsing internal/external webpages, Q&A over csv tables, AWS redshift, data lakes and images, classifying textual and visual information for content filtering and verification.
 - Developed an LLM-guided workflow builder that semantically parses and decomposes complex standard operating procedures (SOPs) into executable DAGs, integrating with orchestration frameworks to streamline automation and accelerate knowledge operations execution in production systems across Amazon worldwide retail stores. **Agents in production, helping business and customer verification, content filtering, returns and reimbursements with more than \$200MM in annualized savings**
 - Led conceptualization and development of **SOP-Bench**, a SOTA RL gym and benchmark to test LLM's reasoning, planning and tool calling capabilities for operationalizing SOPs.
 - Built a **LLM based text-to-sql engine** to power Amazon's business intelligence use cases that supported customers querying an LLM to extract information from relational databases. The system was built on RAGs, Chain-of-Thought (CoT), self-correction, self-consistency techniques and achieved an execution accuracy of **97%** for writing simple SQL queries.
 - Built a **LLM based annotation agentic workflow** that generated annotated training data using a HITL approach for the Prime Air's use cases pertaining to training large scale vision algorithms for Mid-Air Collision Detection Systems. The framework resulted in an 85% efficiency (cost reduction) for generating training data for ground object detection resulting in **savings greater than \$1.7MM**.
- Role **Applied Scientist, Amazon Q Developer (NGDE), Amazon Web Services**
- Work Summary Leading science efforts for the development of Amazon Q Code Developer Agent, Amazon's flagship GenAI product for helping developer generate and transform code using Large Language Models. [Company Announcement](#)

- Led development of code transformation platform that migrated JDK packages saving **4500** developer and **\$260MM**
- Led development of code transformations for packages from AwsSdk v1 to v2, AWS's most used Java SDK platform, reporting updates to **Amazon S-team**

Role Applied Scientist, Artificial General Intelligence Org.

Work Summary Lead Scientist working on fine-tuning, model performance enrichment & model robustness

- Designed a self learning approach that iteratively identifies gaps in model's capabilities and identifies high quality training sets for fine-tuning.
- Invented and deployed a neural network-based model to reduce instability of large language models powering Amazon Alexa by **35%**. **Work published in Association for Computational Linguistic Conference 2023**
- **Built Amazon's core capabilities for Large Language Model's data quality framework** for training large language models (**Amazon Nova Model family**) as well as selection of datasets to bridge model gaps.
- Conceptualized, developed and productionized Amazon Alexa's NLU evaluation framework **to reduce IMR costs by \$5MM.**

Role Applied Scientist, Customer Targeting, WorldWide Retail Stores

Work Summary Conceptualized and deployed Amazon's customer action ranking models as well as MAB-based autonomous targeting systems

- Developed a novel customer ranking algorithm based on neural networks that is currently deployed in all worldwide markets by Amazon's Customer Targeting Team to target millions of retail customers for shopping on Amazon. **Work published in Amazon Machine Learning Conference; A/B testing showed downstream profit of \$250,000 for a period of 2 days in US Marketplace for 10MM customers**
- Developed multi-arm bandit based Objective Based Targeting System that is used by Amazon worldwide to automate marketing campaigns to drive targeting to satisfy a business objective **Deployment showed an increment of 8% across campaigns on electronics, beauty, toys**
- Developed Amazon's Customer Segmentation Insights tool that leverage statistical machine learning to identify key characteristics of customer shopping trends that affect contributing profit.

Previous Affiliations

Jun 2019- Sep 2019	Data Scientist Intern, Amazon.com Inc , World-Wide Amazon Stores Organization, Seattle, USA
August 2016- August 2021	Doctoral Researcher, Washington University in St. Louis , Department of Computer Science & Engineering, St. Louis, USA
Aug 2015- May 2016	Software Engineer, Infosys Limited , Pune, India

Publications

Google Scholar [Arghya Datta](#)

Preprints The following articles are under review and published on *Arxiv*

- Subhrangshu Nandi†, **Arghya Datta†**, Nikhil Vichare†, Indranil Bhattacharya†, Huzefa Raja, Jing Xu, Shayan Ray, Giuseppe Carenini, Abhi Srivastava, Aaron Chan, Man Ho Woo, Amar Kandola, Brandon Theresa, Francesco Carbone *SOP-Bench: Complex Industrial SOPs for Evaluating LLM Agents*, Arxiv, 2025 († denotes equal contribution, co-first authors)

Peer-Reviewed Articles The following articles are published in peer-reviewed premier journal & conferences.

- **Arghya Datta**, Subhrangshu Nandi, Jingcheng Xu, Greg Ver Steeg, He Xie, Anoop Kumar, Aram Galstyan, *Measuring and Mitigating Local Instability in Deep Neural Networks*, Findings for Association of Computational Linguistics (ACL), 2023

- **Arghya Datta**, SJ Swamidass. *Fair-Net: A Network Architecture for Reducing Performance Disparity between Identifiable Sub-populations*, In Proceedings of the 14th International Conference on Agents and Artificial Intelligence (ICAART), 2022
- **Arghya Datta**, Noah R Flynn, SJ Swamidass. *Cal-net: Jointly learning classification and calibration on imbalanced binary classification tasks*, International Joint Conference on Neural Networks (IJCNN), 2021
- **Arghya Datta**, Noah R Flynn, Dustyn A Barnette, Keith F Woeltje, Grover P Miller, S Joshua Swamidass. *Machine learning liver-injuring drug interactions with non-steroidal anti-inflammatory drugs (NSAIDs) from a retrospective electronic health record (EHR) cohort*, PLOS Computational Biology 2021
- Dustyn Barnette, Mary Schleiff, , **Arghya Datta**, Noah Flynn, S Joshua Swamidass, Grover P. Miller. *Meloxicam methyl group determines enzyme specificity for thiazole bioactivation compared to sudoxicam*, Toxicology Letters 2021
- **Arghya Datta**, Matthew K Matlock, Na Le Dang, Thiago Moulin, Keith F. Woeltje, Elizabeth L. Yanik, Sanjay Joshua Swamidass. *Black Box to Conversational Machine Learning: Ondansetron Reduces Risk of Hospital-Acquired Venous Thromboembolism*, IEEE journal of Biomedical and Health Informatics 2020
- Matthew K Matlock, **Arghya Datta**, Na Le Dang, Kevin Jiang, S Joshua Swamidass. *Deep learning long-range information in undirected graphs with wave networks*, International Joint Conference on Neural Networks (IJCNN), 2019

**Amazon
Machine
Learning
Conference
[Internal]**

The following articles are published in peer-reviewed internal Amazon Machine Learning Conference (acceptance rate < 15%), which is attended by top academics from all over the world

- **Arghya Datta**, Pragyana Mishra, *Ranking Customer Actions using Propensity Calibration in Neural Networks*, Amazon Machine Learning Conference (AMLC), 2022
- **Arghya Datta**, Subhrangshu Nandi, Anoop Kumar, Aram Galstyan, Greg Ver Steeg, Abhishek Kumar, Purva Kulkarni. *Measuring and Mitigating Local Instability in Large Language Models*, Amazon Machine Learning Conference (AMLC), 2023

**PhD
Dissertation]**

PhD Thesis defended at Washington University in Saint Louis, U.S.A

- **Datta, Arghya**. Machine Learning in Complex Scientific Domains: Hospitalization Records, Drug Interactions, Predictive Modeling and Fairness for Class Imbalanced Data. Washington University in St. Louis, 2021.

Academic, industry & volunteer service

Peer reviewer

Served as a reviewer for more than **160+** reviews for renowned international Journals and Conferences such as AISTATS, Frontiers in AI, IEEE JBHI, etc.

Research Interests

- Large Language Models (LLMs)
- Agentic AI workflows
- Deep learning
- Knowledge discovery using Machine Learning
- Graph Neural Networks
- Natural Language Understanding
- Time Series Forecasting
- Causal Modelling
- Object detection and Segmentation
- Personalization, Recommendation Systems and Targeting
- Advanced Statistics, A/B testing

Technical Skills

Programming	Python, C, Java, SQL, Bash
LLMs / Generative AI	Pretraining, fine-tuning, and evaluation of Large Language Models (Amazon Titan/Nova models); expertise in prompt engineering (Dspy), retrieval-augmented generation (RAG), knowledge bases, agentic workflows (Langchain, Strands, AgentCore).
Deep Learning	PyTorch (primary), TensorFlow/Keras; experience with model optimization (LoRA, quantization, distillation) and multimodal architectures (vision-language, speech)
NLP / CV	Transformers, Hugging Face ecosystem, spaCy, OpenCV; applied to text generation, summarization, classification, and image processing
Data & Infrastructure	Distributed computing with Spark, Dask, Ray; data pipelines with Parquet, Arrow; experience with vector databases (FAISS, Pinecone, Weaviate) and scalable serving with Triton, vLLM
MLOps / Cloud	Docker, Kubernetes, MLflow, Weights & Biases; deployment on AWS, GCP, Azure; CI/CD pipelines for ML systems
Other Tools	Git, Linux/Unix, REST/gRPC APIs, Pandas, NumPy, JAX (familiar)